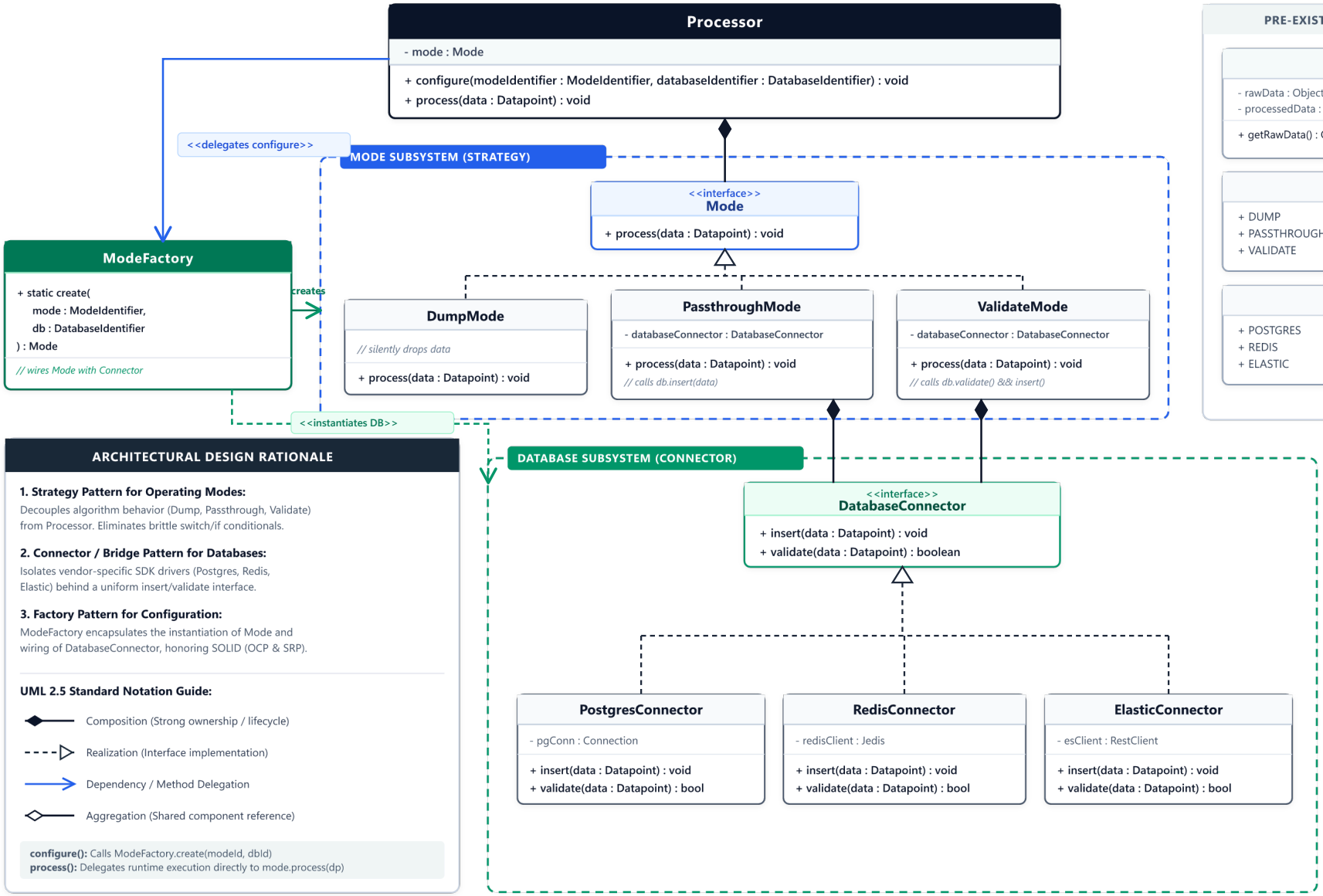
UML CLASS DIAGRAM : PIPELINE DATA PROCESSOR ARCHITECTURE

WALMART GLOBAL TECH SPECIFICATION



PRE-EXISTING PIPELINE TYPES (ASSUMED)

<<abstraction>> Datapoint

- rawData : Object
- processedData : Object

+ getRawData() : Object / setProcessedData()

<<enumeration>> ModelIdentifier

+ DUMP
+ PASSTHROUGH
+ VALIDATE

<<enumeration>> DatabaseIdentifier

+ POSTGRES
+ REDIS
+ ELASTIC

ARCHITECTURAL DESIGN RATIONALE

1. Strategy Pattern for Operating Modes:
Decouples algorithm behavior (Dump, Passthrough, Validate) from Processor. Eliminates brittle switch/if conditionals.

2. Connector / Bridge Pattern for Databases:
Isolates vendor-specific SDK drivers (Postgres, Redis, Elastic) behind a uniform insert/validate interface.

3. Factory Pattern for Configuration:
ModeFactory encapsulates the instantiation of Mode and wiring of DatabaseConnector, honoring SOLID (OCP & SRP).

UML 2.5 Standard Notation Guide:

- Composition (Strong ownership / lifecycle)
- Realization (Interface implementation)
- Dependency / Method Delegation
- Aggregation (Shared component reference)

configure(): Calls ModeFactory.create(modeId, dbId)
process(): Delegates runtime execution directly to mode.process(dp)